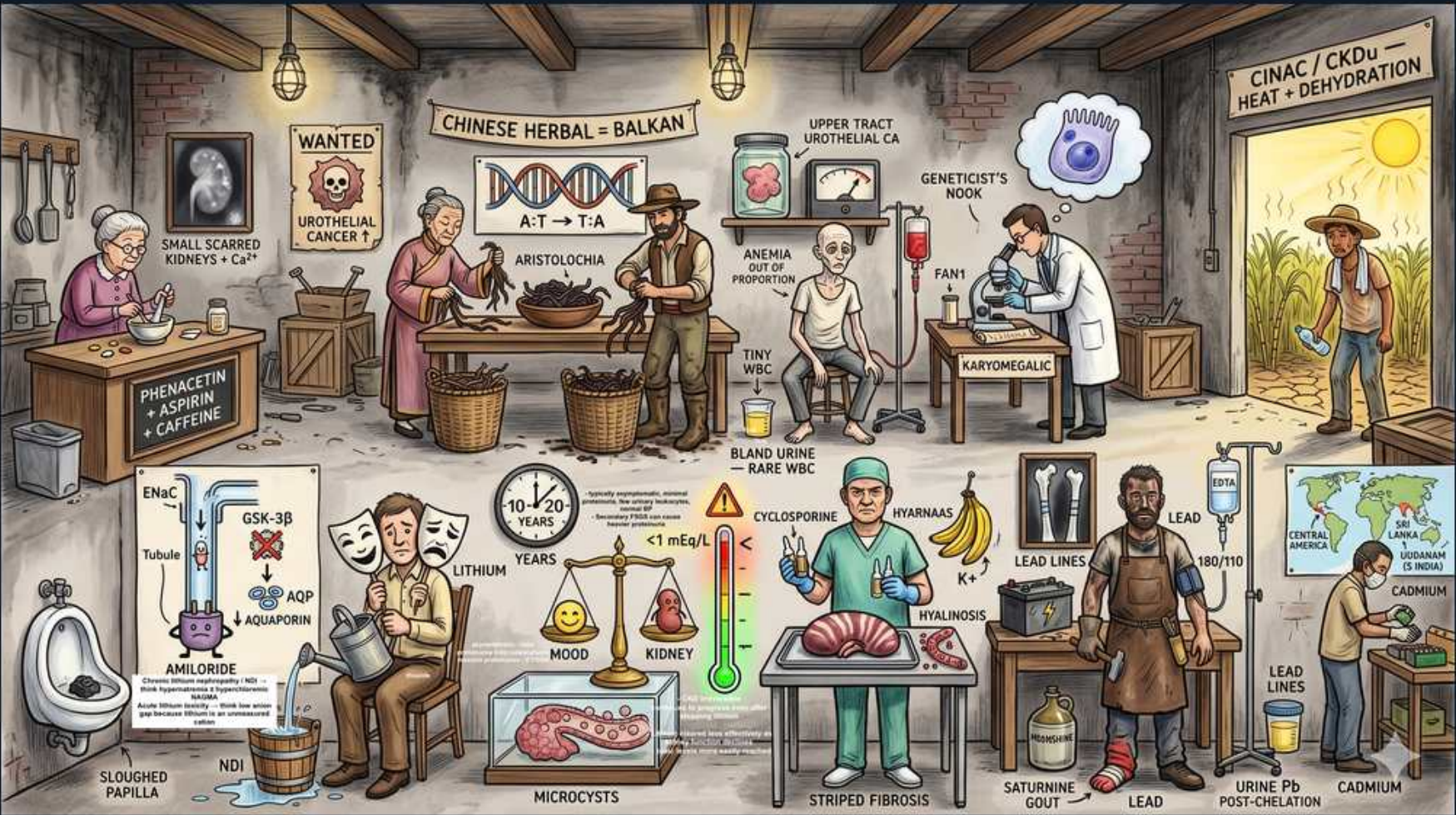


Tubulointerstitial — Toxin chronic CIN



1. Analgesic nephropathy

WHAT YOU SEE

An old woman beside small scarred kidneys marked Ca²⁺ and a PHENACETIN + ASPIRIN + CAFFEINE label — the shrunken bumpy kidneys next to the combination-analgesic mix ARE years of analgesic abuse causing chronic interstitial nephritis.

WHAT IT ENCODES

Analgesic nephropathy is chronic interstitial nephritis from years of combination analgesics — historically PHENACETIN + aspirin + caffeine mixtures, today chronic NSAIDs ± acetaminophen. Mechanism: NSAIDs inhibit medullary vasodilatory prostaglandins → medullary ischemia; phenacetin metabolites cause oxidative papillary injury. Result: small, scarred kidneys with papillary necrosis, sterile pyuria, type I/II RTA, hypertension, and a raised risk of urothelial cancer. Classic patient: middle-aged woman with chronic headaches/back pain and analgesic dependence.

BOARD TRAP

CKD with sterile pyuria and bumpy small kidneys is called 'idiopathic chronic interstitial nephritis' without a drug history. Wrong — a cumulative analgesic exposure history reveals analgesic nephropathy, which is preventable by stopping the analgesics; missing the history loses that lever and the urothelial-cancer surveillance. The medication history is the discriminator.

2. Sloughed papilla

WHAT YOU SEE

A toilet labelled SLOUGHED PAPILLA with an NDI tag — the necrotic papilla passed into the toilet IS papillary necrosis from analgesics, leaving a medulla that can no longer concentrate (nephrogenic DI).

WHAT IT ENCODES

Analgesic nephropathy causes renal papillary necrosis: the medullary papillae infarct and slough into the collecting system, producing gross hematuria, colicky pain (the papilla behaves like a stone), and impaired concentrating ability/nephrogenic DI from medullary destruction. CT urogram shows ring shadows/blunted papillae; sloughed tissue may be passed in urine. The damaged medulla can no longer concentrate, causing polyuria/nocturia.

BOARD TRAP

Painful hematuria with a filling defect in a chronic-analgesic user is treated as a simple ureteral stone. Wrong — the obstructing material is a sloughed necrotic papilla, not a calcium stone; CT urogram showing papillary necrosis and the analgesic history redirect the diagnosis. Papillary necrosis (vs calculus) on imaging is the discriminator.

3. Aristolochic acid / Balkan

WHAT YOU SEE

An aristolochia plant and herbal pot labelled CHINESE HERBAL = BALKAN with an A:T→T:A mutation tag — the slimming herb equated with Balkan grain IS aristolochic acid, and the A:T→T:A transversion IS its signature mutation.

WHAT IT ENCODES

Aristolochic acid (from Aristolochia plants — Chinese herbal weight-loss/slimming remedies; also the cause of Balkan endemic nephropathy via contaminated grain) causes rapidly progressive chronic interstitial fibrosis with hypocellular ('paucicellular') cortical scarring. It forms aristolactam-DNA adducts producing a characteristic A:T→T:A transversion mutation, especially in the TP53 gene, driving carcinogenesis. Often presents as relentless CKD progressing to ESRD even after exposure stops.

BOARD TRAP

A young woman with progressive CKD after taking 'natural/herbal' slimming supplements is reassured the herbs are benign. Wrong — aristolochic-acid-containing herbs cause irreversible fibrosing nephropathy and urothelial cancer; the exposure must be stopped and cancer surveillance started. The herbal-supplement history (often omitted as 'natural') is the discriminator.

4. Urothelial cancer risk

WHAT YOU SEE

A WANTED poster reading UROTHELIAL CANCER with a skull — the most-wanted skull IS the high lifetime risk of upper-tract urothelial carcinoma from aristolochic acid.

WHAT IT ENCODES

Aristolochic acid is a potent carcinogen of the urothelium → high lifetime risk of UPPER-TRACT urothelial carcinoma (renal pelvis/ureter), and also bladder cancer; the A:T→T:A signature mutation is the molecular fingerprint. Patients with aristolochic/Balkan nephropathy or chronic analgesic nephropathy need long-term urothelial cancer surveillance (urine cytology, imaging) because tumors are often multifocal.

BOARD TRAP

Hematuria in a patient with aristolochic-acid or analgesic nephropathy is attributed solely to the papillary necrosis/chronic disease and not investigated for malignancy. Wrong — these patients have a markedly elevated urothelial cancer risk, so new hematuria mandates a cancer workup (cytology, cystoscopy, upper-tract imaging). The carcinogen exposure is the discriminator that makes hematuria a cancer red flag.

5. Anemia out of proportion

WHAT YOU SEE

A pale, gaunt man labelled ANEMIA OUT OF PROPORTION with a tiny WBC icon — his anemia far worse than his modest renal failure IS the clue pointing to lead or aristolochic toxicity rather than plain CKD anemia.

WHAT IT ENCODES

Anemia disproportionately severe relative to the degree of renal impairment is a classic clue in certain toxin nephropathies — notably aristolochic acid and lead nephropathy — reflecting bone-marrow/erythropoietin effects beyond simple CKD. (Lead also causes basophilic stippling and sideroblastic-type anemia.) When the hemoglobin is far lower than the modest creatinine would predict, broaden the differential toward these toxins.

BOARD TRAP

Anemia far out of proportion to a mild creatinine elevation is dismissed as 'expected anemia of CKD.' Wrong — anemia of CKD tracks with GFR; disproportionate anemia points toward lead toxicity (stippling, elevated blood lead) or aristolochic acid, or a hematologic process. The mismatch between anemia severity and renal function is the discriminator that should prompt a toxin/heme workup.

6. Lithium nephropathy

WHAT YOU SEE

A principal-cell tubule with an ENaC channel, GSK-3 β tag and an AMILORIDE block, beside mood masks and a LITHIUM label — lithium sneaking in through ENaC (blocked by amiloride) IS the cause of its nephrogenic DI.

WHAT IT ENCODES

Lithium enters principal cells of the collecting duct through the epithelial sodium channel (ENaC), accumulates, inhibits aquaporin-2 signaling (via GSK-3 β) and causes acquired nephrogenic diabetes insipidus (polyuria/polydipsia) plus, over years, chronic interstitial nephritis with microcysts. AMILORIDE blocks ENaC and reduces lithium uptake, treating the lithium-induced nephrogenic DI. Maintain therapeutic levels (~0.6–1.2 mEq/L), avoid dehydration/NSAIDs/thiazides which raise lithium.

BOARD TRAP

Lithium-induced nephrogenic DI is treated by adding a THIAZIDE alone, or polyuria is treated with desmopressin. Wrong — AMILORIDE (ENaC blocker) is the targeted therapy that reduces lithium entry; thiazides can raise serum lithium (volume contraction → reabsorption) risking toxicity, and DDAVP fails because the defect is nephrogenic. Amiloride vs thiazide/DDAVP is the discriminator.

7. Lithium microcysts + slow course

WHAT YOU SEE

A MICROCYSTS label with a 10-20 YEARS scale weighing mood against kidney — the tiny cysts and decade timeline ARE lithium's slow, duration-dependent CKD, balanced against its psychiatric benefit.

WHAT IT ENCODES

Chronic lithium therapy causes small renal microcysts (seen on MRI/ultrasound) and a slow decline in GFR over 10–20 years; the renal injury is dose- and duration-dependent and may progress even after lithium is stopped. Management is a risk-benefit balance: lithium is highly effective for bipolar disorder, so monitor renal function and consider alternatives only if CKD progresses, with psychiatry co-management.

BOARD TRAP

Lithium is reflexively and abruptly stopped at the first sign of any creatinine change. Wrong — for stable mild changes, lithium's psychiatric benefit may outweigh slow renal risk and abrupt discontinuation risks dangerous mood relapse; the decision requires monitoring and shared decision-making with psychiatry, not knee-jerk cessation. The benefit-risk balance (not automatic discontinuation) is the discriminator.

8. Cyclosporine / CNI toxicity

WHAT YOU SEE

A surgeon labelled CYCLOSPORINE beside STRIPED FIBROSIS, arteriolar HYALINOSIS and a K+-up tag — the banded fibrosis and nodular hyaline arterioles ARE the histologic fingerprint of calcineurin-inhibitor toxicity (improves with dose reduction, unlike rejection).

WHAT IT ENCODES

Calcineurin inhibitors (cyclosporine, tacrolimus) cause both acute (afferent arteriolar vasoconstriction → reversible hemodynamic GFR drop) and chronic nephrotoxicity: striped interstitial fibrosis, nodular arteriolar HYALINOSIS, and tubular atrophy. They also cause hyperkalemia (type IV-like RTA), hypertension, hypomagnesemia, and (cyclosporine more than tacrolimus) gingival hyperplasia/hirsutism. Drug levels and many CYP3A4 interactions matter; nephrotoxicity is dose-related.

BOARD TRAP

In a transplant patient with rising creatinine on a calcineurin inhibitor, the answer assumes acute rejection and escalates immunosuppression. Wrong — CNI nephrotoxicity mimics rejection but improves with DOSE REDUCTION, whereas rejection needs more immunosuppression — opposite management. Drug level and biopsy (arteriolar hyalinosis/striped fibrosis for CNI toxicity vs tubulitis/C4d for rejection) is the discriminator.

9. Lead nephropathy

WHAT YOU SEE

A worker labelled LEAD with gingival LEAD LINES and a SATURNINE GOUT tag — the blue gum lines and gouty joint ARE chronic lead nephropathy impairing urate excretion.

WHAT IT ENCODES

Chronic lead exposure (old paint, batteries, contaminated moonshine, industrial work) causes chronic interstitial nephritis with proximal tubular dysfunction; systemic clues include 'lead lines' (gingival Burton lines and dense metaphyseal lines on x-ray), hypertension, microcytic anemia with basophilic stippling, abdominal colic, peripheral neuropathy (wrist drop), and saturnine gout. Lead impairs uric acid excretion → hyperuricemia and gout.

BOARD TRAP

Hyperuricemia/gout plus CKD in a patient with an industrial or moonshine exposure history is treated only as 'primary gout.' Wrong — the picture is saturnine gout from lead nephropathy; the workup needs a blood lead level and EDTA mobilization test, and treatment includes chelation, not just urate-lowering. The lead exposure history (and stippling/lead lines) is the discriminator.

10. Saturnine gout / urine Pb

WHAT YOU SEE

A SATURNINE GOUT label with a URINE Pb POST-CHELATION sample — the lead measured in urine after an EDTA challenge IS the mobilization test that reveals the stored bone-lead body burden a single blood level can miss.

WHAT IT ENCODES

Saturnine gout = gout caused by lead-induced reduction in renal uric acid clearance. Lead nephropathy is confirmed by an elevated blood lead level and, classically, the EDTA (calcium disodium edetate) lead-mobilization test, which measures urinary lead excretion after a chelation challenge — high mobilizable lead confirms a significant body burden. Treatment is removal from exposure plus chelation (EDTA, succimer/DMSA) for symptomatic/high-burden cases.

BOARD TRAP

Lead nephropathy is excluded based on a single near-normal blood lead level. Wrong — a normal current blood lead does not exclude a high cumulative bone-lead burden from remote exposure; the EDTA chelation-mobilization (urine lead after challenge) detects the stored body burden. The mobilization test vs a one-time blood level is the discriminator.

11. Cadmium

WHAT YOU SEE

A CADMIUM worker beside an itai-itai / map icon — cadmium accumulating in the proximal tubule IS Fanconi-type LMW (β 2-microglobulin) proteinuria plus painful osteomalacia.

WHAT IT ENCODES

Chronic cadmium exposure (smelting, battery/pigment manufacture, contaminated rice/soil — 'itai-itai' disease in Japan) accumulates in the proximal tubule causing proximal tubular dysfunction/Fanconi syndrome (low-molecular-weight proteinuria with β 2-microglobulin, glucosuria, aminoaciduria), chronic interstitial nephritis, and osteomalacia/osteoporosis (bone pain — 'itai-itai' = 'ouch-ouch'). Urinary β 2-microglobulin is an early biomarker.

BOARD TRAP

Low-molecular-weight (tubular) proteinuria in a cadmium-exposed worker is dismissed because the standard ALBUMIN dipstick is negative. Wrong — cadmium causes LMW tubular proteinuria (β 2-microglobulin) that the albumin dipstick misses; you must measure tubular proteins/ β 2-microglobulin. The dipstick-negative LMW proteinuria is the discriminator that should prompt tubular-protein testing.

12. CINAC / CKDu — heat + dehydration

WHAT YOU SEE

A laborer in a sugarcane field under a blazing sun labelled CINAC / CKDu HEAT + DEHYDRATION — the young worker baking in the heat IS Mesoamerican nephropathy: CKD from recurrent heat stress without diabetes or hypertension.

WHAT IT ENCODES

Chronic interstitial nephritis in agricultural communities (CINAC) / CKD of unknown origin (CKDu) — also called Mesoamerican nephropathy — affects young male manual laborers (sugarcane, rice) in hot climates with recurrent heat stress, volume depletion and possibly agrochemical/heavy-metal co-exposure. Features: slowly progressive CKD with bland urine, hyperuricemia, hypokalemia, and small kidneys, WITHOUT the usual diabetes/hypertension drivers. Prevention is hydration and rest/shade breaks.

BOARD TRAP

Progressive CKD in a young Central American agricultural worker is attributed to diabetic or hypertensive nephropathy. Wrong — CINAC/CKDu occurs WITHOUT diabetes or significant hypertension, in young laborers with recurrent heat/dehydration and bland urine; the demographic and absence of classic drivers is the discriminator from diabetic/hypertensive CKD.

13. Karyomegalic nephropathy

WHAT YOU SEE

A geneticist at a microscope labelled KARYOMEGALIC with a FAN1 tag and a giant-nucleus cell — the strikingly enlarged tubular nuclei plus FAN1 ARE the rare genetic karyomegalic interstitial nephritis (not cancer or virus).

WHAT IT ENCODES

Karyomegalic interstitial nephritis is a rare GENETIC chronic interstitial nephritis caused by biallelic FAN1 mutations (a DNA interstrand-crosslink repair gene), producing tubular epithelial cells with strikingly enlarged, hyperchromatic nuclei (karyomegaly) plus interstitial fibrosis and progressive CKD in young adults. It may follow respiratory infections and there can be a family history; diagnosis is by biopsy showing karyomegaly and confirmed genetically.

BOARD TRAP

Karyomegalic tubular cells with enlarged atypical nuclei on biopsy are interpreted as 'malignancy/viral cytopathic change.' Wrong — in young adults with familial CIN this is the benign genetic karyomegalic interstitial nephritis (FAN1), not cancer or a virus; recognizing the genetic entity avoids an oncologic workup and prompts family genetic counseling. The genetic/familial context and FAN1 testing is the discriminator.